

End-to-End MERN Application Architecture & Best Practices

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GITHUB LINK FOR LAB: <https://github.com/kaiserfranxx/MERN-Application-lab>

OVERVIEW:

This project serves as a demonstration of a basic implementation of MERN (MongoDB, Express.js, React, Node.js) application. The goal of this lab activity was to better understand how the frontend and backend of applications communicate, implement CRUD operations, and follow project structuring.

The application is very simple, allowing for users to add and view items stored in a MongoDB database through a React-based UI.

TECHNOLOGIES USED IN LAB:

- Node.js
- Express.js
- MongoDB w/ Mongoose
- React.js
- Axios
- dotenv
- cors

APPLICATION STRUCTURE

The application is divided into two main parts:

- **BACKEND**

- Handles the server logic and API routes.
- Connects to MongoDB.
- Uses Mongoose for data modeling purposes.

- **FRONTEND**

- Built via React.
- Fetches and displays data from the backend API.
- Makes use of Axios for HTTP requests.

HOW THE APPLICATION WORKS:

1. The React frontend sends a request to the backend API.
2. The Express server receives the request that was sent.
3. The server interacts with MongoDB through the use of Mongoose.
4. The response (data) is sent back towards to the frontend.
5. The frontend updates and displays the data that was sent.

FEATURES IMPLEMENTED IN APPLICATION:

- View all items (GET request)
- Add new items (POST request)
- Fetch item by ID (GET request)
- Display items dynamically through React

CHALLENGES DURING LAB ACTIVITY:

1. CORS Errors - The frontend was initially unable to communicate with the backend due to CORS restrictions. Was solved by enabling the CORS middleware by adding:
`app.use(cors());`
2. Misspellings - Misspelled several essential components within the backend like `ItemsList()` that prevented me from running the backend initially. Was solved by correcting such mistakes and properly connecting them to their proper form.

OUTPUT SCREENSHOT:

